

Bidirectional Wireless Bridge: S7-200 PLC to ESP32

This document details the architecture and the specific "Heartbeat" timing logic for a bidirectional communication system between an industrial S7-200 PLC and a wireless ESP32-S3 node.

1. System Components & Connectivity

A. The PLC Station (S7-200 CPU 224XP)

- **Physical Button (I0.0):** Local input. Triggers wireless LEDs.
- **Physical Bulb (Q0.0):** Local output. Triggered by wireless buttons.
- **Communication:** Port 0 (RS-485) connected to the MAX485 module.

B. The Gateway (ESP32)

- **Role:** Modbus RTU Slave (Address 1). It translates electrical RS-485 signals into 2.4GHz wireless data.
- **Mirroring:** It maintains a "Virtual Register Table" in its RAM (Addresses 40001 and 40002).

2. The Logic Flow: From Wire to Wireless

Scenario A: Remote Sensor -> Local PLC Bulb

1. **Wireless Event:** You press a button on the **Remote ESP32**. It sends an ESP-NOW packet: MSG:BTN_1:1.
2. **Gateway Receipt:** The **Gateway ESP32** receives the packet and writes 1 into its internal Modbus Register **40001**.
3. **The Polling Cycle:** The PLC (acting as Master) sends a request: *"Hey Slave 1, what is in Register 40001?"*
4. **Data Transfer:** The Gateway replies: 01 03 02 00 01 79 84.
5. **PLC Action:** The PLC places that 1 into its internal memory VW100. The logic VW100 == 1 turns on the physical bulb Q0.0.

Scenario B: Local PLC Button -> Remote Wireless LED

1. **Local Event:** You press the PLC button **I0.0**.
2. **Logic Update:** The PLC moves the value 1 into its local memory VW102.
3. **The Push Cycle:** During the next communication cycle, the PLC sends a "Write" command to the Gateway: *"Set Register 40002 to 1."*
4. **Gateway Trigger:** The Gateway notices **40002** has changed. It immediately sends an ESP-NOW packet: CMD:LED:1.
5. **Remote Action:** The **Remote ESP32** receives the text and turns on its physical LED.

3. The "Self-Resetting" Timer (T37)

Why we chose this specific timer type

In the S7-200, we used a **Self-Resetting Pulse Generator** (using a Normally Closed contact of the timer itself). We chose this because Modbus is a **Half-Duplex** protocol (only one device can talk at a time).

A. Preventing Communication "Flooding"

The PLC CPU scans its logic every 1-5 milliseconds. If we triggered the Modbus block without a timer, the PLC would try to send 500 requests per second.

- **Result without Timer:** The RS-485 bus would "crash" because the Gateway wouldn't have finished responding to the first request before the second one arrived.
- **Result with T37:** By setting PT: 5 (500ms), we limit the "conversation" to twice per second. This is the "Sweet Spot" for responsiveness without overloading the hardware.

B. The Need for a "Rising Edge"

The MBUS_MSG instruction requires a **0 to 1 transition** on the First pin to execute.

- If you used a standard switch, you would have to flip it manually every time you wanted an update.
- The T37 pulse creates a "High" signal for exactly **one scan cycle** and then drops to "Low." This automatic "flicker" ensures the Modbus block sees a fresh start every 0.5 seconds.

Logic Sample (LAD vs Chronogram)

LAD: [T37 NC]----(TON T37, 5)

Chronogram:

Power: _____

T37: 
^ 500ms ^ 500ms

4. Why Modbus RTU is Used

For this project, we utilize Modbus RTU because it is the industry standard for serial communication. Unlike proprietary protocols, it is highly compatible with generic microcontrollers like the ESP32.

1. **Standardization:** Modbus RTU is the "Universal Language" of automation. By using Modbus, your ESP32 gateway can talk to almost any PLC in the world (Schneider, Delta, Mitsubishi), not just Siemens.
2. **Simplicity:** It uses a straightforward request-response model that is easy to implement in

both PLC Ladder Logic and C++ for the ESP32.

3. **Efficiency:** It requires minimal overhead, making it ideal for the limited processing power of legacy PLCs while maintaining reliable data integrity through CRC checks.

5. Summary Table

Feature	Description	Importance
Baud Rate	9600	Matches ESP32 Serial speed.
Timer T37	500ms Pulse	Prevents data collisions.
VW100/102	V-Memory	Acts as the "Post Office" for your data.
MAX485	RS-485 Chip	Converts 5V TTL logic to Differential Signal for long distance.